

Human Review: Decomposable-space tube construction

Original automatic proof: `solution_packet.pdf`

Checked date: 14 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears to be correct. Although it does not solve the full problem for arbitrary infinite-dimensional Banach spaces, I would regard the argument as both quite ingenious and quite general. The only structural assumption is a decomposition

$$X = Y \oplus Z$$

with both Y and Z infinite-dimensional. Since Banach spaces which do not admit such decompositions are rather special and exotic from the usual fixed-point-theoretic point of view, this covers a large and natural class of spaces.

Idea of the Construction

The proof uses the decomposition $X = Y \oplus Z$ as a system of base and fibre coordinates. The Y -coordinate is first retracted onto a small sphere

$$aS_Y.$$

Over each base point $y \in aS_Y$, the construction places a small fibre in the Z -direction of radius

$$r(y) > 0.$$

The radius function is Lipschitz, positive everywhere, and has infimum zero. This is the essential device: positivity prevents fixed points, while the vanishing infimum gives null minimal displacement.

The map then acts by flipping the sign in the Z -fibre:

$$y + r(y)z \mapsto y - r(y)z.$$

After the first iterate, the base point and radius are fixed, and the dynamics alternate between the two fibre signs. This gives

$$T^2 = S, \quad T^3 = T,$$

so the iterates are uniformly Lipschitz.

Technical Points Checked

The auxiliary estimates appear sound. The radial clipping map is 2-Lipschitz, and the variable-radius sphere selection lemma gives the needed Lipschitz control for

$$H(r, q) = rR_Z(\kappa_Z(q/r)).$$

This is what ensures that the fibre selection remains Lipschitz even though the radius varies with the base point.

The use of infinite-dimensionality is also clear. It is needed to invoke the Benyamini–Sternfeld Lipschitz retractions

$$R_Y : B_Y \rightarrow S_Y, \quad R_Z : B_Z \rightarrow S_Z,$$

and to build a positive Lipschitz gauge on S_Y whose infimum is zero.

Review Outcome

Archive this packet as an apparently correct partial solution to Q2. It gives an affirmative answer for every decomposable Banach space with two infinite-dimensional complemented summands. It should not be described as a full solution to the arbitrary Banach-space problem, but it is a strong and conceptually useful partial result.